| Question | | | | Marking details | Marks Available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **8 FT 2 HT** | (a) | | | Correct addition shown or by implication (1)<br>Conclusion must match addition (1) | | | 2 | 2 | 1 | |
| | (b) | (i) | | Shows an <u>increase</u> [Jan-May], reaches a <u>constant</u> [May-Jul] and <u>decreases</u> [Jul-Dec]<br>**Alternative** for reaches a <u>constant</u> [May-Jul] **–** reaches a maximum for the summer months<br>**Don't accept** same shape | | | 1 | 1 | 1 | |
| | | (ii) | | Biased information / information misleading or unfair / maximum values are misleading<br>**Don't accept** different amounts of sunshine each year | | | 1 | 1 | | |
| | (c) | (i) | | $2^{nd}$, $5^{th}$ and $6^{th}$ boxes ticked i.e.<br>Apr 16 was the month that had the most actual sunshine hours (1)<br>Apr16 had 4 times the actual number of sunshine hours compared to Jan 16 (1)<br>There is only one month where the actual and the expected sunshine hours were the same (1)<br>Lose 1 mark for each extra tick | | | 3 | 3 | | |
| | | (ii) | | $\frac{600}{200}$ = 3 [kW] **or** $\frac{600}{3}$ = 200 [h] **or** 200 × 3 = 600 [kWh] (1)<br>Claim [is correct] must be consistent and backed up by calculations (1) | | | 2 | 2 | 1 | |
| | (d) | (i) | | 3 670 – 3 400 = 270 [kWh] (1)<br>270 × 29 = 7 830 [p] (1)<br>**Alternative:**<br>3 670 × 29 = 106 430 [p] **and** 3 400 × 29 = 98 600 [p] (1)<br>106 430 [p] - 98 600 [p] = 7 830 [p] (1)<br>Don't award the answer mark for £78.30 p | | 2 | | 2 | 2 | |
| | | (ii) | | Reduce [payback] time / quicker [payback] time | 1 | | | 1 | | |
| | | | | **Question 8 FT**<br>**Question 2 HT total** | **1** | **2** | **9** | **12** | **5** | **0** |